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1 字符串

1.1 Manacher

```
1 void manacher(int n, char *s, int *d) {
2     int l = 0, r = -1;
3     For(i, 1, n) {
4         int &k = d[i];
5         k = i > r ? 1 : min(d[l + r - i], r - i + 1);
6         while (s[i - k] == s[i + k]) k++;
7         if (i + k - 1 > r) r = i + k - 1, l = i - k + 1;
8     }
9 }
```

1.2 Z 函数

```
1 void zfunc(int n, const int *const s, int *const z) {
2     fill(z + 1, z + n + 1, 0), z[1] = n;
3     int l = 0;
4     For(i, 2, n) {
5         if (z[l] + l - 1 >= i) z[i] = min(z[i - l + 1], l + z[l] - i);
6         while (s[z[i] + 1] == s[i + z[i]]) z[i]++;
7         if (i + z[i] > l + z[l]) l = i;
8     }
9 }
```

1.3 最小表示法

```
1 int minrep(auto &a) {
2     int n = a.size(), i = 0, j = 1, k = 0;
3     while(i < n && j < n && k < n) {
4         if(a[(i + k) % n] == a[(j + k) % n]) {
5             ++k;
6         } else {
7             if(a[(i + k) % n] > a[(j + k) % n]) swap(i, j);
8             j += k + 1, j += i == j, k = 0;
9         }
10    }
11    return i;
12 }
13 // rotate(a.begin(), a.begin() + minrep(a), a.end());
```

1.4 SA

```
1 const int N = 1e6 + 10;
2 int sa[N], rk[N << 1], st[20][N];
3 void SA(int n, auto *a) {
4     static int b[N], c[N];
5     int m = *max_element(a + 1, a + n + 1);
6     For(i, 1, n) b[i] = i, rk[i] = a[i];
7     For(i, n + 1, n * 2) rk[i] = 0;
8     auto bs = [&]() {
9         memset(c, 0, m + 1 << 2);
```

```
10         For(i, 1, n) ++c[rk[i]];
11         For(i, 1, m) c[i] += c[i - 1];
12         rFor(i, n, 1) sa[c[rk[b[i]]]--] = b[i];
13     };
14     bs();
15     for (int t = 1; t < n; t += t) {
16         int tot = 0;
17         For(i, n - t + 1, n) b[++tot] = i;
18         For(i, 1, n) if(sa[i] > t) b[++tot] = sa[i] - t;
19         bs();
20         b[sa[1]] = m = 1;
21         For(i, 2, n) {
22             int x = sa[i], y = sa[i - 1];
23             b[x] = m += (rk[x] != rk[y] || rk[x + t] != rk[y + t]);
24         }
25         memcpy(rk, b, n + 1 << 2);
26         if(m == n) break;
27     }
28     int k = 0;
29     a[n + 1] = -1;
30     For(i, 1, n) {
31         k -= k > 0;
32         while (a[i + k] == a[sa[rk[i] - 1] + k]) ++k;
33         st[0][rk[i]] = k;
34     }
35     For(i, 1, __lg(n)) For(j, 1, n - (1 << i) + 1) {
36         st[i][j] = min(st[i - 1][j], st[i - 1][j + (1 << i - 1)]);
37     }
38 }
39 int lcp(int x, int y) {
40     x = rk[x], y = rk[y];
41     if(x > y) swap(x, y);
42     int k = __lg(y - x);
43     return min(st[k][x + 1], st[k][y - (1 << k) + 1]);
44 }
```

2 数据结构

2.1 李超树

```
1 namespace DS {
2     const int S = 1e7 + 10;
3     pll tr[S];
4     int ls[S], rs[S], tot;
5     ll eval(pll x, ll y) {
6         return x[0] * y + x[1];
7     }
8     void bd() {
9         memset(ls, 0, sizeof(ls)), memset(rs, 0, sizeof(rs));
10        tot = 0;
11    }
12    void ins(pll x, int &k, int l = 1, int r = 1e6) {
```

```

13     if(!k) return tr[k = ++tot] = x, void();
14     int m = l + r >> 1, fl;
15     if(eval(tr[k], m) > eval(x, m)) swap(x, tr[k]);
16     if((fl = eval(tr[k], l) <= eval(x, l)) && eval(tr[k], r) <= eval(x, r)) return;
17     !fl ? ins(x, ls[k], l, m) : ins(x, rs[k], m + 1, r);
18 }
19 ll que(int x, int k, int l = 1, int r = 1e6) {
20     if(!k) return Z;
21     int m = l + r >> 1;
22     return min(eval(tr[k], x), x <= m ? que(x, ls[k], l, m) : que(x, rs[k], m + 1,
23         r));
24 } // namespace DS

```

2.2 全局平衡二叉树

树上点权修改 + 查询最大权独立集数值, 每次修改 $O(\log n)$ ——先全局平衡化建结构, 之后点改权更新。

- 完整程序(自带 main): 输入 n, m (点权重与操作数); 每行操作 p, y , 且 p 需要先异或上一次答案(在线防作弊), 改点权后输出当前答案
- 依赖: 无(自带 mat 2×2 max-plus 矩阵、与 $N = 10^6$ 的全局数组)
- 实现: 重链重度剖分 + 树剖点上矩乘, 矩阵 que 计算链顶的独立集
- 初始化: $dfs0$ (轻重儿子与父亲)、 $dfs1$ (矩阵的初始 f)、 $build$ (按点权重分极值点) 全部在 $main$ 内调好

坑: 输入 $p \hat{=} ans$ 的在线加密别忘了; $f[i][1]$ 初始即点权 $a[i]$; 改权用 $upd(p, y - a[p])$ 差值更新, 和 $a[p] = y$ 同步, 整棵树必须从 1 号点开始 DFS (树无向)。

```

1  const int N = 1e6 + 10;
2  const int INF = 1e9;
3  struct mat {
4      int a[2][2];
5      int *operator[](int k) { return a[k]; }
6      const int *operator[](int k) const { return a[k]; }
7      mat operator*(const mat &b) const {
8          return {
9              max(a[0][0] + b[0][0], a[0][1] + b[1][0]),
10             max(a[0][0] + b[0][1], a[0][1] + b[1][1]),
11             max(a[1][0] + b[0][0], a[1][1] + b[1][0]),
12             max(a[1][0] + b[0][1], a[1][1] + b[1][1]);
13         }
14     };
15 int n;
16 vector<int> t[N];
17 int sz[N], s[N], tp[N], dw[N], fa[N];
18 void dfs0(int u) {
19     sz[u] = 1;
20     for(auto v : t[u]) {
21         fa[v] = u;
22         t[v].erase(find(all(t[v]), u));
23         dfs0(v);

```

```

24         if(sz[v] > sz[s[u]]) s[u] = v;
25         sz[u] += sz[v];
26     }
27     if(s[u]) t[u].erase(find(all(t[u]), s[u]));
28 }
29 int dfn[N], wt[N], rt[N];
30 int f[N][2];
31 pll dfs1(int u) {
32     static int dt;
33     dfn[u] = ++dt;
34     wt[dt] = sz[u] - sz[s[u]];
35     if(!tp[u]) tp[u] = u;
36     if(!s[u]) return dw[tp[u]] = u, pll{f[u][0], f[u][1]};
37     tp[s[u]] = tp[u];
38     pll sf = dfs1(s[u]);
39     for(auto v : t[u]) {
40         auto [x, y] = dfs1(v);
41         f[u][0] += max(x, y);
42         f[u][1] += x;
43     }
44     return {f[u][0] + max(sf[0], sf[1]), f[u][1] + sf[0]};
45 }
46 struct tree_t {
47     int ls, rs;
48     mat x;
49 } tr[N << 1];
50 #define ls (tr[k].ls)
51 #define rs (tr[k].rs)
52 int build(int l, int r) {
53     if(l == r) return l << 1;
54     int m, s1 = 0, s2 = 0;
55     for(i, l, r) s1 += wt[i];
56     for(i, l, r) if((s2 += wt[i]) * 2 > s1) {
57         m = i;
58         break;
59     }
60     m -= m == r;
61     return tr[m << 1 | 1] = {build(l, m), build(m + 1, r)}, m << 1 | 1;
62 }
63 void pushup(int k) { tr[k].x = tr[rs].x * tr[ls].x; }
64 void upd(int p, mat x, int k, int l, int r) {
65     if(l == r) return tr[k].x = x, void();
66     int m = k >> 1;
67     p <= m ? upd(p, x, ls, l, m) : upd(p, x, rs, m + 1, r);
68     pushup(k);
69 }
70 mat que(int L, int R, int k, int l, int r) {
71     if(L <= l && r <= R) return tr[k].x;
72     int m = l + r >> 1;
73     if(R <= m) return que(L, R, ls, l, m);
74     if(m < L) return que(L, R, rs, m + 1, r);
75     return que(L, R, rs, m + 1, r) * que(L, R, ls, l, m);
76 }

```

```

77 #undef ls
78 #undef rs
79 void apply(int p) {
80     upd(dfn[p], {f[p][0], f[p][1], f[p][0], -INF}, rt[tp[p]], dfn[tp[p]],
        dfn[dw[tp[p]]]);
81 }
82 pll que(int p) {
83     auto res = que(dfn[p], dfn[dw[tp[p]]], rt[tp[p]], dfn[tp[p]], dfn[dw[tp[p]]]);
84     return {max(res[0][0], res[1][0]), max(res[0][1], res[1][1])};
85 }
86 void upd(int p, int dt) {
87     f[p][1] += dt;
88     auto u = [&](int k, int w) {
89         if(!k) return;
90         auto [x, y] = que(k);
91         f[fa[k]][0] += w * max(x, y);
92         f[fa[k]][1] += w * x;
93     };
94     for(; p; p = fa[tp[p]]) {
95         u(tp[p], -1);
96         apply(p);
97         u(tp[p], 1);
98     }
99 }
100 int getans() {
101     auto [x, y] = que(1);
102     return max(x, y);
103 }
104 int a[N];
105 int main() {
106     int m;
107     cin >> n >> m;
108     For(i, 1, n) cin >> a[i], f[i][1] = a[i];
109     For(i, 1, n - 1) {
110         int u, v;
111         cin >> u >> v;
112         t[u].push_back(v);
113         t[v].push_back(u);
114     }
115     dfs0(1), dfs1(1);
116     For(i, 1, n) if(tp[i] == i) rt[i] = build(dfn[tp[i]], dfn[dw[tp[i]]]);
117     For(i, 1, n) apply(i);
118     int ans = 0;
119     For(i, 1, m) {
120         int p, y;
121         cin >> p >> y;
122         p ^= ans;
123         upd(p, y - a[p]), a[p] = y;
124         ans = getans(), cout << ans << endl;
125     }
126 }

```

2.3 LCT

$O(\log n)$ 摊还的动态 lct:支持 link/cut/路径聚合(默认异或,改 pu 可换求和或最值)。

- init(): 先 n 与 $a[N]$ 就位,然后对 $1..n$ 初始化
- link(u, v): 连边(自动判环);cut(u, v): 断边;que(u, v): 路径聚合值;upd(u, x): 单点改值并维护
- mkr/access/splay/fd 为内部机制,一般不用管
- 依赖: 全局 N 与点权数组 $a[N]$

坑: 若改 pu 换成加法/最值,注意初始值(0/INF 类)与空链的传递;cut 仅当边存在时删(发现 $fa[v] == u$ 才断);路径查询前一定要 mkr 换根。

```

1 namespace LCT {
2     #define ls(k) ch[k][0]
3     #define rs(k) ch[k][1]
4     int ch[N][2], fa[N];
5     bool lz[N];
6     int tr[N];
7     inline void pu(int k) {
8         tr[k] = tr[ls(k)] ^ tr[rs(k)] ^ a[k];
9     }
10    inline void pl(int k) {
11        if(k) lz[k] ^= 1, swap(ls(k), rs(k));
12    }
13    inline void pd(int k) {
14        if(lz[k]) pl(ls(k)), pl(rs(k)), lz[k] = 0;
15    }
16    inline bool gc(int k) { return ch[fa[k]][1] == k; }
17    inline bool ir(int k) { return ch[fa[k]][gc(k)] != k; }
18    void pda(int k) {
19        static int st[N], t;
20        for(st[++t] = k; !ir(k);) st[++t] = k = fa[k];
21        while(t) pd(st[t--]);
22    }
23    void rot(int u) {
24        int f = fa[u], &g = fa[f], t = gc(u), x = ch[u][!t];
25        if(!ir(f)) ch[g][gc(f)] = u;
26        fa[u] = g, g = u;
27        ch[u][!t] = f, ch[f][t] = x;
28        if(x) fa[x] = f;
29        pu(f);
30    }
31    void splay(int u) {
32        for(pda(u); !ir(u); rot(u))
33            if(!ir(fa[u])) {
34                rot(gc(u) == gc(fa[u]) ? fa[u] : u);
35            }
36        pu(u);
37    }
38    void access(int u0) {
39        for(int u = u0, p = 0; u; p = u, u = fa[u]) splay(u), rs(u) = p, pu(u);
40        splay(u0);
41    }

```

```

42 int fd(int u) {
43     for(access(u); ls(u); u = ls(u), pd(u)) ;
44     return splay(u), u;
45 }
46 void mkr(int u) {
47     access(u), pl(u);
48 }
49 void init() {
50     For(i, 1, n) pu(i);
51 }
52 void link(int u, int v) {
53     mkr(u);
54     if(fd(v) != u) mkr(v), fa[v] = u;
55 }
56 void cut(int u, int v) {
57     mkr(u), access(v), access(u);
58     if(fa[v] == u) fa[v] = 0;
59 }
60 void upd(int u, int x) {
61     a[u] = x, splay(u);
62 }
63 int que(int u, int v) {
64     mkr(u), access(v);
65     return tr[v];
66 }
67 } // namespace LCT

```

2.4 广义串并联图

```

1 struct SPQR_tree {
2     int m = 1, fa[N];
3     bool typ[N];
4     unordered_map<int, int> t[N];
5     bitset<N> vis;
6     int node(int u, int v, bool t) {
7         tp[+m] = t, fa[u] = t, fa[v] = t;
8         return m;
9     }
10    void add(int u, int v, int w) {
11        if(t[u].count(v)) {
12            t[u][v] = t[v][u] = node(t[u][v], w, 0);
13        } else {
14            t[u][v] = t[v][u] = w;
15        }
16    }
17    void build(int n) {
18        queue<int> q;
19        auto ins = [&](int u) { if (!vis[u] && t[u].size() <= 2) vis[u] = 1,
20        q.push(u); };
21        For(i, 1, n) ins(i);
22        while(q.size()) {
23            int u = q.front();

```

```

23        q.pop();
24        if(t[u].empty()) continue;
25        if(t[u].size() == 1) {
26            int v = t[u].begin()->first;
27            t[v].erase(u), ins(v);
28        } else {
29            int x = t[u].begin()->first, y = next(t[u].begin())->first;
30            int w = node(t[u][x], t[u][y], 1);
31            t[x].erase(u), t[y].erase(u);
32            add(x, y, w);
33            ins(x), ins(y);
34        }
35    }
36 }
37 } dl;
38 int *fa = dl.fa, &m = dl.m;
39 bool *typ = dl.typ;

```

2.5 二分栈

固定罚分下做“分段 DP 并提供前驱/段数”的斜率优化, $O(n \log n)$ 。

- work(dt): 每段代价 calc(j+1, i) 减去罚分 dt, 求 f[n](值 + 段数, nd{cnt}); 返回 nd{f.x = 数值, f.c = 段数}
- 数据结构: 单调栈 + 二分区间: q 维护尾点范围, get(p, L) 比较新旧决策
- 依赖: 全局 N(数组) 与 calc(l, r) 函数
- 通常配 WQS 使用: 外层二分罚分, 这里给出对应的最优段数

坑: 要求 calc 满足四边形不等式(决策单调), 否则二分栈回退不正确; nd 的 operator< 是 (x, c) 字典序, 保证匹配时优先小罚分; calc 是 $O(1)$ 还是 $O(n)$ 需注意, 总复杂度将乘上它。

```

1 struct nd {
2     ll x;
3     int c;
4     bool operator<(const nd b) const {
5         return x < b.x || (x == b.x && c < b.c);
6     }
7     nd operator+(const nd b) const { return {x + b.x, c + b.c}; }
8 };
9 nd work(ll dt) {
10    static nd f[N];
11    deque<pii> q;
12    f[0] = {0, 0}, q.push_back({n, 0});
13    auto get = [&](int j, int i) -> nd {
14        return f[j] + nd{calc(j + 1, i) - dt, 1};
15    };
16    For(i, 1, n) {
17        f[i] = get(q.front()[1], i);
18        if(q.front()[0] <= i) q.pop_front();
19        while(q.size()) {
20            auto [R, p] = q.back(); q.pop_back();
21            int L = q.size() ? q.back()[0] + 1 : i + 1;

```

```

22         if(!(get(p, L) < get(i, L))) continue;
23         int l = L, r = R + 1;
24         while(l + 1 < r) {
25             int m = l + r >> 1;
26             get(p, m) < get(i, m) ? l = m : r = m;
27         }
28         q.push_back({l, p});
29         break;
30     }
31     q.push_back({n, i});
32 }
33 return f[n];
34 }

```

2.6 wqs 二分构造方案

外层 WQS 二分会惩罚值,内层调用黑盒 Wilber(区间 DP 的裸段数相关结果),最终返回分割点数组。

- 接口: `wqs(n, k, que, up)`; `que(l, r)` 为区间代价函数(返回 `i128`); `up` 为代价上界(二分范围 `-up..1`)
- 返回: 0-based 分割点数组(以 0 起、`n` 终),恰含 k 个区间段
- 依赖: `Wilber` 类必须事先提供(该文件未给出)——通常是带决策单调性的最优区间划分解
- 复杂度: $O(\log C)$ 次内层 `Wilber`

坑: WQS 二分要求「段数更多的代价序列」满足凸性(否则失效);内部 `que` 每段自动减去二分惩罚 `dt`, `que` 必须对任意 l, r 可控区间; `fp/gp` 数组拼结细节在 `get`, 若边界出错多为 `Wilber` 允许的段数与返回不符。

```

1 vect<int> wqs(int n, int k, auto &&que, i128 up) {
2     int B = __lg(n) + 1, T = (1 << B) - 1;
3     auto chk = [&](i128 dt) {
4         return Wilber<i128>(
5             n, [&](int l, int r) { return (que(l, r) - dt) << B | 1; }, 1e36);
6     };
7     auto get = [&](i128 dt) {
8         auto [f, fp] = chk(dt);
9         vect<int> pt;
10        for (int i = n; i; i = fp[i]) pt += i;
11        pt += 0;
12        reverse(all(pt));
13        return pt;
14    };
15    i128 l = -up - 1, r = 1;
16    while (l + 1 < r) {
17        i128 m = l + r >> 1;
18        (chk(m).first[n] & T) < k ? (l = m) : (r = m);
19    }
20    auto fp = get(l), gp = get(l + 1);
21    if (gp.size() - 1 == k) return gp;
22    int j = 1;
23    For(i, 1, gp.size() - 2) {
24        while (fp[j] < gp[i]) ++j;
25        if (fp[j - 1] <= gp[i - 1] && fp.size() - j + i - 1 == k) {

```

```

26            return gp.substr(0, i) + fp.substr(j);
27        }
28    }
29    assert(0);
30 }

```

2.7 $O(1)$ RMQ

线性预处理后任意区间最值 $O(1)$ (常数很小,可自定义比较器)。

- 模板参数: `rmq_t<T, cmp, dt>`; `cmp` 视作“取最优比较器”(less<T> 取 max, greater<T> 取 min); `dt` 是下标偏移(默认 1 即 1-based 输入与查询)
- 接口: `bd(n, T* a) / bd(vector<T>&) / bd(n, func)(func(i) 从偏移处取值)`, 然后 `que(l, r)`
- 复杂度: 建 $O(n)$, 查 $O(1)$
- 依赖: 无(自带模板参数)

坑: `st` 分块表仅对块间查询, `dt` 默认为 1 但 `bd(vector)` 传 `data()` - `dt` 指针, 注意边界; `cmp` 需要严格弱序; 长查询数据为 n 时块数 $(n-1)/B$ 别超 `st` 层数(已按 `__lg` 计算)。

```

1 template <typename T, typename cmp = less<T>, int dt = 1>
2 struct rmq_t {
3     static constexpr int B = 20;
4     vector<T> a, pre, suf;
5     vector<vector<T>> st;
6     vector<int> f;
7     T _que(int L, int R) {
8         int k = __lg(R - L + 1);
9         return max(st[k][L], st[k][R - (1 << k) + 1], cmp());
10    }
11    T _que_bitc(int k, int L, int R) {
12        int l = k * B, x = f[R] & ~((1 << L - l) - 1);
13        return a[l + __lg(x & -x)];
14    }
15    void bd(int n, T *a) {
16        a.resize(n), pre.resize(n), suf.resize(n), f.resize(n);
17        For(i, 0, n - 1) pre[i] = suf[i] = a[i] = _a[i + dt];
18        int k = (n - 1) / B;
19        st.resize(__lg(k + 1) + 1), st[0].resize(k + 1);
20        For(i, 0, k) {
21            int l = i * B, r = min(i * B + B - 1, n - 1);
22            rFor(i, r - 1, l) pre[i] = max(pre[i], pre[i + 1], cmp());
23            For(i, l + 1, r) suf[i] = max(suf[i], suf[i - 1], cmp());
24            st[0][i] = suf[r];
25            static int b[B + 1];
26            int t = 0, s = 0;
27            For(i, l, r) {
28                while(t && cmp()(a[b[t]], a[i])) s ^= 1 << b[t--] - l;
29                f[i] = s ^= 1 << i - l;
30                b[++t] = i;
31            }
32        }
33        For(i, 1, __lg(k + 1)) {

```

```

34         int m = k - (1 << i) + 1;
35         st[i].resize(m + 1);
36         For(j, 0, m) st[i][j] = max(st[i - 1][j], st[i - 1][j + (1 << i - 1)],
cmp());
37     }
38 }
39 void bd(vector<T> &a) { bd(a.size(), a.data() - dt); }
40 void bd(int n, auto func) {
41     vector<T> a(n);
42     For(i, 0, n - 1) a[i] = func(i + dt);
43     bd(a);
44 }
45 T que(int l, int r) {
46     l -= dt, r -= dt;
47     int il = l / B, ir = r / B;
48     if(il == ir) {
49         return _que_bitc(il, l, r);
50     } else {
51         T s = max(pre[l], suf[r], cmp());
52         if(il + 1 < ir) s = max(s, _que(il + 1, ir - 1), cmp());
53         return s;
54     }
55 }
56 };

```

3 图论

3.1 网络流

Dinic 最大流,附赠残余网络割集。

- add(u, v, w) 加边; solve(s, t, n) 传点数并返回最大流; cut(f) 后 $f[i] = \frac{0}{1}$ 标记 $\frac{s}{t}$ 侧
- 边数上限 $N = 10^6$ (静态数组), 无穷大 $Z = 1e18$
- 复杂度 $O(n^2 m)$ 上界, 单位容量图 $O(m \sqrt{n})$

坑: bfs() 首行 $d[i] = -(i == s)$ 是笔误——只有 s 得 -1, 其余 0, 而 $!d[v]$ 只认 -1 当未访问, 导致 BFS 进不了任何点, 最大流恒为 0 (已实测)。改成 $d[i] = -1$ 即可。

```

1 class Dinic_t {
2     public:
3         static constexpr int N = 1e6 + 10;
4         static constexpr ll Z = 1e18;
5         struct edge {
6             int v, n;
7             ll w;
8         } e[N];
9         int n, hd[N], cur[N], d[N], tot = 1, s, t;
10        void _add(int u, int v, ll w) {
11            e[++tot] = {v, hd[u], w}, hd[u] = tot;
12        }
13        void add(int u, int v, ll w) {
14            _add(u, v, w), _add(v, u, 0);
15        }
16    };

```

```

15 }
16 void clear() {
17     For(i, 1, n) hd[i] = 0;
18     n = s = t = 0, tot = 1;
19 }
20 bool bfs() {
21     For(i, 1, n) d[i] = -(i == s), cur[i] = hd[i];
22     queue<int> q;
23     q.push(s);
24     while (q.size()) {
25         int u = q.front();
26         q.pop();
27         if (u == t) return 1;
28         for (int i = cur[u]; i; i = e[i].n)
29             if (e[i].w && !d[e[i].v]) {
30                 d[e[i].v] = d[u] + 1, q.push(e[i].v);
31             }
32     }
33     return 0;
34 }
35 ll dfs(int u, ll in) {
36     if (u == t) return in;
37     ll out = 0;
38     for (int &i = cur[u]; i; i = e[i].n)
39         if (e[i].w && d[e[i].v] == d[u] + 1) {
40             ll s = dfs(e[i].v, min(in, e[i].w));
41             e[i].w -= s, e[i ^ 1].w += s;
42             in -= s, out += s;
43             if (!in) break;
44         }
45     if (!out) d[u] = -1;
46     return out;
47 }
48 ll solve(int _s, int _t, int _n) {
49     s = _s, t = _t, n = _n;
50     ll ans = 0;
51     while (bfs()) ans += dfs(s, Z);
52     return ans;
53 }
54 void cut(auto *f) {
55     For(i, 1, n) f[i] = 1;
56     queue<int> q;
57     f[s] = 0;
58     q.push(s);
59     while (q.size()) {
60         int u = q.front();
61         q.pop();
62         for (int i = hd[u]; i; i = e[i].n)
63             if (e[i].w && f[e[i].v]) {
64                 f[e[i].v] = 0;
65                 q.push(e[i].v);
66             }
67     }

```

```
68     }
69 } f;
```

3.2 费用流

$O(f \cdot n \log m)$ 的势优化费用流,以及能处理初始负环/下界的 mcmf2。

- `add(u, v, w, c)` 加容量 w 费用 c 的边;mcmf(s, t) 返回 {流量, 费用};mcf = true 时费用为负(增广更优)才继续
- `mcmf2(s, t)`: 先消除初始负费用边(翻转并按差额补平衡边),适合有负环或下界流约束的图;临时辅助边用完自动回收
- 依赖: 无:容量流 f 由输入决定

坑: mcmf() 的初始势靠 SPFA,原图有负环时报错/死循环,只能用 mcmf2;mcf=1 表示“任意流中最小费用”而非固定流量。

```
1 struct MCMF_t {
2     static constexpr int N = 1e6 + 10, FL = 1e9;
3     static constexpr ll Z = 1e18;
4     struct edge {
5         int v, n, w;
6         ll c;
7     } e[N];
8     int n, hd[N], tot = 1, s, t, p[N];
9     void _add(int u, int v, int w, ll c) {
10         e[++tot] = {v, hd[u], w, c}, hd[u] = tot;
11     }
12     void add(int u, int v, int w, ll c) {
13         _add(u, v, w, c), _add(v, u, 0, -c);
14     }
15     void clear() {
16         For(i, 1, n) hd[i] = 0;
17         tot = 1, n = s = t = 0;
18     }
19     ll h[N], d[N];
20     void spfa() {
21         static bool vis[N];
22         For(i, 1, n) h[i] = Z;
23         queue<int> q;
24         q.push(s), h[s] = 0, vis[s] = 1;
25         while(q.size()) {
26             int u = q.front(), v;
27             q.pop();
28             vis[u] = 0;
29             for(int i = hd[u]; i; i = e[i].n)
30                 if(e[i].w && h[u] + e[i].c < h[v = e[i].v]) {
31                     h[v] = h[u] + e[i].c;
32                     if(!vis[v]) q.push(v), vis[v] = 1;
33                 }
34         }
35     }
36     bool dij() {
```

```
37         For(i, 1, n) d[i] = Z;
38         priority_queue<pair<ll, int>> q;
39         q.emplace(d[s] = 0, s);
40         int u;
41         while(q.size()) {
42             auto [tmp, u] = q.top();
43             q.pop();
44             if(-tmp != d[u]) continue;
45             for(int i = hd[u]; i; i = e[i].n)
46                 if(e[i].w) {
47                     int v = e[i].v;
48                     auto w = d[u] + h[u] - h[v] + e[i].c;
49                     if(w < d[v]) d[v] = w, p[v] = i, q.emplace(-w, v);
50                 }
51         }
52         For(i, 1, n) if(d[i] != Z) h[i] = d[i] += h[i];
53         return d[t] != Z;
54     }
55     pair<int, ll> mcmf(int _s, int _t, bool mcf = 0, int _n = 0) {
56         s = _s, t = _t, n = _n;
57         spfa();
58         int a1 = 0;
59         ll a2 = 0;
60         while(dij() && (!mcf || d[t] < 0)) {
61             int m = FL;
62             for(int u = t; u != s; u = e[p[u] ^ 1].v) m = min(m, e[p[u]].w);
63             a1 += m, a2 += m * d[t];
64             for(int u = t; u != s; u = e[p[u] ^ 1].v) e[p[u]].w -= m, e[p[u] ^ 1].w
65                 += m;
66             return {a1, a2};
67         }
68         pair<int, ll> mcmf2(int _s, int _t, bool mcf = 0, int _n = 0) {
69             s = _s, t = _t, n = _n;
70             static int d[N];
71             int a1 = 0;
72             ll a2 = 0;
73             For(i, 2, tot) if(i % 2 == 0 && e[i].c < 0) {
74                 int u = e[i ^ 1].v, v = e[i].v;
75                 swap(e[i].w, e[i ^ 1].w);
76                 d[u] -= e[i ^ 1].w, d[v] += e[i ^ 1].w;
77                 a2 += e[i ^ 1].w * e[i].c;
78             }
79             For(i, 1, _n) {
80                 d[i] > 0 ? add(_n + 1, i, d[i], 0) : add(i, _n + 2, -d[i], 0);
81             }
82             add(_t, _s, FL, 0);
83             a1 += e[tot].w, a2 += mcmf(_n + 1, _n + 2).second;
84             tot -= 2 * (_n + 1);
85             For(i, 1, _n) while(hd[i] > tot) hd[i] = e[hd[i]].n;
86             auto [a3, a4] = mcmf(_s, _t, mcf);
87             return {a1 + a3, a2 + a4};
88         }
```

```
89 };
```

3.3 支配树

$O((n + m) \alpha)$ 求有向图从根 1 出发的支配树。

- `add(u, v)` 加有向边;`getdom(n)` 一次求出,`dm[u]` 为 u 的直接支配点(根为 0)
- 内部 `dfn` 是 DFS 序;`sd` 半支配点
- 复杂度接近线性,常数适中, $N = 10^6$

坑: 根固定为 1,图里所有点必须从 1 可达,否则未达点 `dm` 无意义;`que()` 是递归路径压缩,深图栈压力大时可改迭代。

```
1 namespace DOM {
2
3 const int N = 1e6 + 10;
4 int n;
5 vect<int> t[N], t2[N], q1[N];
6 int fa[N], f1[N], dfn[N], pos[N], dt, sd[N], dm[N];
7 pii f[N];
8 void add(int u, int v) {
9     t[u] += v, t2[v] += u;
10 }
11 void dfs(int u) {
12     pos[dfn[u] = ++dt] = u;
13     for(auto v : t[u])
14         if(!dfn[v]) {
15             dfs(v), f1[v] = u;
16         }
17 }
18 pii que(int u) {
19     if(fa[u] == u) return f[u];
20     cmin(f[u], que(fa[u]));
21     fa[u] = fa[fa[u]];
22     return f[u];
23 }
24 void getdom(int _n) {
25     n = _n, dfs(1);
26     For(i, 1, n) fa[i] = i, f[i] = {dfn[i]};
27     rFor(i, dt, 1) {
28         int u = pos[i], s = i;
29         for(auto x : q1[u]) dm[x] = que(x)[1];
30         if(i == 1) break;
31         for(auto x : t2[u]) cmin(s, que(x)[0]);
32         sd[u] = pos[s];
33         q1[sd[u]] += u;
34         f[u] = {dfn[sd[u]], u};
35         for(auto v : t[u])
36             if(f1[v] == u) fa[v] = u;
37     }
38     For(i, 2, dt) {
39         int u = pos[i];
40         if(sd[dm[u]] == sd[u])
```

```
41         dm[u] = sd[u];
42     else
43         dm[u] = dm[dm[u]];
44     }
45 }
46
47 } // namespace DOM
```

3.4 一般图最大匹配

```
1 int n, fa[N];
2 vect<int> t[N];
3 int fd(int u) { return fa[u] == u ? u : fa[u] = fd(fa[u]); }
4 void getmat(int rt) {
5     static int vis[N], mat[N], pr[N];
6     For(i, 1, n) vis[i] = pr[i] = 0, fa[i] = i;
7     queue<int> q;
8     auto lca = [](int x, int y) {
9         static int vis[N], t; ++t;
10        while(1) {
11            if(x && exchange(vis[x = fd(x)], t) == t) return x;
12            x = pr[mat[x]], swap(x, y);
13        }
14        return -1;
15    };
16    auto blossom = [&](int x, int y, int z) {
17        for(; fd(x) != z; x = pr[y]) {
18            pr[x] = y, y = mat[x], fa[x] = fa[y] = z;
19            if(vis[y] == 1) vis[y] = 2, q.push(y);
20        }
21    };
22    vis[rt] = 2, q.push(rt);
23    while(q.size()) {
24        int u = q.front(); q.pop();
25        for(auto v : t[u]) {
26            if(!vis[v]) pr[v] = u, vis[v] = 1, vis[mat[v]] = 2, q.push(mat[v]);
27            if(!mat[v]) {
28                while(v) mat[v] = pr[v], swap(mat[pr[v]], v);
29                return;
30            }
31            if(vis[v] == 2) {
32                int l = lca(u, v);
33                blossom(u, v, l), blossom(v, u, l);
34            }
35        }
36    }
37 }
```

4 数学

4.1 Barret 约简

```
1 struct barret {
2     u128 w; u64 M;
3     barret(const u64 M) : M(M), w(u64((u128(1) << 64) / M)) {}
4     inline u64 mod(u64 x) const {
5         x -= ((w * x) >> 64) * M;
6         return x >= M ? x - M : x;
7     }
8     inline u64 mod_without_chk(u64 x) const { return x - ((w * x) >> 64) * M; }
9 };
```

4.2 拉格朗日插值

已知 $n+1$ 个点 $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$, 其中 x_i 互不相同, 则存在唯一的次数不超过 n 的多项式 $L(x)$ 满足 $L(x_i) = y_i$.

Lagrange 插值多项式为:

$$L(x) = \sum_{i=0}^n y_i \prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j}$$

当 x_i 是连续整数 (如 $0, 1, \dots, n$) 时, 可 $O(n)$ 计算 $L(x)$ (需预处理阶乘逆元)。此时:

$$L(x) = \sum_{i=0}^n y_i * \frac{\prod_{j=0, j \neq i}^n (x - j)}{(-1)^{n-i} i! (n-i)!}$$

其中分子可通过前后缀积 $O(1)$ 得到。

```
1 vector<mint> lag(vector<pair<mint, mint>> a) {
2     int n = a.size();
3     vector<mint> f(n + 1), ans(n + 1, 0);
4     f[0] = 1;
5     For(i, 1, n) {
6         auto [x, y] = a[i - 1];
7         rFor(j, i - 1, 0) f[j + 1] += f[j], f[j] *= -x;
8     }
9     For(i, 1, n) {
10        auto [x, y] = a[i - 1];
11        mint w = 1, inv = !x ? 0 : -x.inv(), z = 0;
12        For(j, 1, n) if(i != j) w *= x - a[j - 1].first;
13        w = w.inv() * y;
14        if(!x) For(j, 1, n) ans[j - 1] += f[j] * w;
15        else For(j, 0, n) z = (f[j] - z) * inv, ans[j] += z * w;
16    }
17    return ans;
18 }
19
20 // 需要 fac(n), ifac(n)
21 void lag_i(int n, mint *a) {
22     vector<mint> f(n + 1, 0), g(n, 0);
```

```
23     f[0] = 1;
24     For(i, 1, n) rFor(j, i - 1, 0) f[j + 1] += f[j], f[j] *= mint::raw(mint::getM()
25     - i);
26     For(i, 1, n) {
27         mint inv = -ifac(i) * fac(i - 1), w = a[i] * ifac(n - i) * ifac(i - 1) * ((i
28         ^ n) & 1 ? -1 : 1), s = 0;
29         For(j, 0, n - 1) s = (f[j] - s) * inv, g[j] += s * w;
30     }
31 }
32 mint lag_p(int n, mint *a, mint x) {
33     if(x.val() < n) return a[x.val()];
34     vector<mint> f(n + 1), g(n + 1);
35     mint ans = 0;
36     f[0] = g[n] = 1;
37     For(i, 0, n - 1) f[i + 1] = f[i] * (x - i);
38     rFor(i, n - 1, 0) g[i] = g[i + 1] * (x - i);
39     For(i, 0, n - 1) {
40         mint w = f[i] * g[i + 1] * a[i] * ifac(n - 1 - i) * ifac(i) * ((i ^ n) & 1 ?
41         1 : -1);
42         ans += w;
43     }
44     return ans;
```

4.3 Pollard Rho

```
1 constexpr bool chkp(ll n) {
2     constexpr int p[] = {2, 325, 9375, 28178, 450775, 9780504, 1795265022}; // {2,
3     auto ksm = [&](i128 a, ll b) {
4         i128 s = 1;
5         for(; b >>= 1, a = a * a % n) b & 1 && (s = s * a % n, 0);
6         return s;
7     };
8     if(n <= 40)
9         return binary_search(p, p + sizeof(p), n);
10    else {
11        if(n % 2 == 0) return 0;
12        int t = __builtin_ctzll(n - 1), i = 0;
13        for(auto b : p) {
14            i128 x = ksm(b, n - 1 >> t);
15            if(x == 1) continue;
16            for(i = 0; i < t && x != n - 1; ++i) x = x * x % n;
17            if(i == t) return 0;
18        }
19        return 1;
20    }
21 }
22 ll PR(ll n) {
23     mt19937_64 rng(114514);
24     if(n % 2 == 0) return 2;
```

```

25 static constexpr int S = 127;
26 uniform_int_distribution<> rnd(1, n - 1);
27 ll x = 0, y = 0, c = rnd(rng), w = 1, g;
28 auto f = [&](ll x) { return (i128(x) * x + c) % n; };
29 for(int t = 1;; ++t) {
30     x = f(x), y = f(f(y));
31     if(x == y) {
32         x = y = 0, c = rnd(rng);
33         continue;
34     }
35     w = i128(w) * abs(x - y) % n;
36     if(!w) return __gcd(abs(x - y), n);
37     if(!(y & S) && (g = __gcd(w, n)) > 1) return g;
38 }
39 }
40 vect<ll> fact(ll n) {
41     if(n == 1) return {};
42     vect<ll> f;
43     auto S = [&](ll n, auto &&S) {
44         if(chkp(n)) return f += n, void();
45         ll x = PR(n);
46         while(n % x == 0) n /= x;
47         S(x, S);
48         if(n > 1) S(n, S);
49     };
50     S(n, S);
51     sort(all(f)), f.erase(unique(all(f)), f.end());
52     return f;
53 }

```

4.4 min_25

$O(N^{0.75} / \log N)$ 求积性函数 f 的前缀和,要求 f 在素数点 $f(p)$ 是关于 p 的低次多项式(最后拆成 p 的幂的线性组合)。

- 接口: solve(n),内部依次 init()(线性筛出 10^5 内质数与 $s_1/s_2/s: \Sigma p, \Sigma p^2, \Sigma(p^2-p)$)+ solveg(n) (求 g 数组)+ $f(n, m)$ 递归求和
- func(x): 素数幂 $x = p^k$ 处 f 的取值,题变则改(示例 $x(x-1)$ 即 $f(p) = p(p-1)$,欧拉函数类)
- 上限: $\text{sqrt}(n) \leq 10^5$,即 $n \leq 10^{10}$;模数固定 $M = 1e9 + 7$
- g 数组大小 $2e5$ (双值映射 $\text{id}(n)$)

坑: solve 内部已调 init(),重复使用只需再次 solve;sqrt(n) 为整平方时内部 fl 有特殊处理,别乱改;数组 g_1/g_2 是模 M 下的量。

```

1 const int M = 1e9 + 7;
2 constexpr int i2 = M + 1 >> 2, i6 = (M + 1) / 6;
3 void add(int &a, int b) {
4     a += b;
5     a >= M && (a -= M);
6 }
7 void sub(int &a, int b) {
8     a -= b;

```

```

9     a < 0 && (a += M);
10 }
11 const int N = 1e5 + 10;
12 int p[N], pc, p2[N];
13 int s1[N], s2[N], s[N];
14 void init() {
15     static const int n = 1e5;
16     static int b[N];
17     For(i, 2, n) {
18         if(!b[i]) p[++pc] = i;
19         For(j, 1, pc) if(i * p[j] <= n) {
20             b[i * p[j]] = 1;
21             if(i % p[j] == 0) break;
22         }
23         else break;
24     }
25     p[pc + 1] = n + 1;
26     For(i, 1, pc) {
27         p2[i] = 1ll * p[i] * p[i] % M;
28         s1[i] = (s1[i - 1] + p[i]) % M;
29         s2[i] = (s2[i - 1] + p2[i]) % M;
30         s[i] = (s2[i] - s1[i] + M) % M;
31     }
32 }
33 ll lim, n0;
34 int g1[N * 2], g2[N * 2];
35 int id(ll n) {
36     return n <= lim ? n : n0 / n + lim;
37 }
38 void solveg(ll _n0) {
39     n0 = _n0;
40     lim = sqrt(n0);
41     ll fl = n0 / lim == lim;
42     auto init = [&](ll n, int _id) {
43         n %= M;
44         sub(g1[_id] = n * (n + 1) / 2 % M, 1);
45         sub(g2[_id] = n * (n + 1) % M * (n + n + 1) % M * i6 % M, 1);
46     };
47     auto doit = [&](ll n, int m, int i1, int i2) {
48         sub(g1[i1], 1ll * (g1[i2] - s1[m - 1] + M) * p[m] % M);
49         sub(g2[i1], 1ll * (g2[i2] - s2[m - 1] + M) * p2[m] % M);
50     };
51     For(i, 1, lim) init(i, i);
52     For(i, 1, lim - fl) init(n0 / i, i + lim);
53     For(i, 1, pc) {
54         ll x = p[i], y = x * x, z = n0 / x;
55         For(j, 1, min(lim - fl, n0 / y)) doit(n0 / j, i, j + lim, id(z / j));
56         rFor(j, lim, min(lim + 1, y)) doit(j, i, j, j / x);
57     }
58 }
59 int g(ll n) { return (g2[id(n)] - g1[id(n)] + M) % M; }
60 ll func(ll n) {
61     n %= M;

```

```

62     return n * (n - 1) % M;
63 }
64 int f(ll n, int m) {
65     if(p[m] > n) return 0;
66     ll ans = g(n) - s[m] + M;
67     For(j, m + 1, pc) {
68         int p = ::p[j];
69         if(1ll * p * p > n) break;
70         for(ll s = p; s * p <= n; s *= p) {
71             ans += func(s) * f(n / s, j) % M + func(s * p);
72         }
73     }
74     return ans % M;
75 }
76 ll solve(ll n) {
77     init(), solveg(n);
78     return (f(n, 0) + 1) % M;
79 }

```

4.5 类欧

$O(\log n)$ 求 $\sum_{i=0}^n \lfloor (a \cdot i + b) / c \rfloor$ 及其平方、带权。

- 结构版 nd f(int n, int a, int b, int c): 参数顺序 (n, a, b, c) , 一次递归同时得到 $\{f, g, h\}$:
 $\triangleright f = \sum \lfloor (ai+b)/c \rfloor; g = \sum \lfloor \cdot \rfloor^2; h = \sum i \cdot \lfloor (ai+b)/c \rfloor$
- 单值版 f(int a, int b, int c, int n): 参数顺序 (a, b, c, n) , 只求 f
- 依赖: mint(带 inv 等)、全局 i2/i6

坑: 两个 f 重载签名都是 (int, int, int, int), C++ 按返回类型不计入签名, 不能同时编译——实际用哪版复制哪版, 并留心两版参数顺序相反; 递归深度 $O(\log n)$, 无栈风险。

```

1 const mint i2 = mint(2).inv(), i6 = mint(6).inv();
2 struct nd {
3     mint f, g, h;
4 };
5 nd f(int n_, int a, int b, int c) {
6     mint n = n_, x = a / c, y = b / c, s0 = n + 1, s1 = n * (n + 1) * i2, s2 = n *
    (n + 1) * (n + n + 1) * i6;
7     if(!a) return {s0 * y, s0 * y * y, s1 * y};
8     if(a >= c || b >= c) {
9         nd rs = f(n_, a % c, b % c, c);
10        mint f = rs.f, g = rs.g, h = rs.h;
11        return {f + s1 * x + s0 * y, g + s2 * x * x + s0 * y * y + (x * h + y * f +
    s1 * x * y) * 2, h + s2 * x + s1 * y};
12    }
13    int m = (1ll * a * n_ + b) / c;
14    nd rs = f(m - 1, c, c - b - 1, a);
15    mint f = rs.f, g = rs.g, h = rs.h;
16    return {n * m - f, n * m * m - h * 2 - f, i2 * (n * n * m + n * m - g - f)};
17 }
18 mint f(int a, int b, int c, int n) {
19     if(!a) return mint(b / c) * (n + 1);
20     if(a >= c || b >= c) {

```

```

21     mint x;
22     if(n & 1)
23         x = mint(n + 1 >> 1) * n * (a / c);
24     else
25         x = mint(n >> 1) * (n + 1) * (a / c);
26     return f(a % c, b % c, c, n) + x + mint(n + 1) * (b / c);
27 }
28 int m = (1ll * a * n + b) / c;
29 return mint(n) * m - f(c, c - b - 1, a, m - 1);
30 }

```

5 数论

5.1 分数还原

$q \equiv \frac{x}{a} \pmod{p}$, $x \leq A$, $|a|$ 取到最小值

```

1 pii approx(int x, int M, int A) {
2     int y = M, a = 1, b = 0;
3     while(x > A) {
4         swap(x, y), swap(a, b), a -= x / y * b, x *= y;
5     }
6     return {x, a};
7 }

```

5.2 $O(V)$ - $O(1)$ GCD

```

1 struct fgcd_t {
2     static constexpr int n = 1e7, B = 3163;
3     array<int, 3> a[n + 1];
4     int s[B + 1][B + 1];
5     bitset<n + 1> b;
6     fgcd_t() {
7         vect<int> p;
8         a[1] = {1, 1, 1};
9         For(i, 2, n) {
10             if(!b[i]) i <= B && (p += i, 0), a[i] = {i, 1, 1};
11             for(auto z : p) {
12                 int x = i * z; if(x > n) break;
13                 b[x] = 1, a[x] = a[i];
14                 *min_element(all(a[x])) *= z;
15                 if(i % z == 0) break;
16             }
17         }
18         For(i, 0, B) For(j, 0, i) s[i][j] = !j ? i : s[j][i % j];
19     }
20     int operator()(int x, int y) const {
21         if(!x || !y) return x | y;
22         if(x == 1 || y == 1) return 1;
23         auto g1 = [&](int x, int y) {
24             int t;
25             if(y == 1 || (t = x % y) == 0) return y;

```

```

26         return y > B ? 1 : s[y][t];
27     };
28     int a0 = g1(x, a[y][0]), a1 = g1(x /= a0, a[y][1]), a2 = g1(x /= a1, a[y]
[2]);
29     return a0 * a1 * a2;
30 }
31 };

```

5.3 离散对数

```

1 ll BSGS(ll a, ll b, ll p) {
2     a %= p, b %= p;
3     if(p == 1) return 0;
4     if(b == 1) return 0; // a^0 = 1
5     if(!a) return -1;
6     // l = a 的阶: l | p - 1, 从 p - 1 出发按素因子反复试除
7     ll l = p - 1;
8     for(ll q = 2; q * q <= l; ++q)
9         for(; l % q == 0 && ksm(a, l / q, p) == 1; l /= q);
10    for(; l > 1 && ksm(a, l - 1, p) == 1; l = 1; // 剩下的素因子最多一个
11    // m = ceil(sqrt(l)), 保证 m * m >= l; 大解落在 (m, m * m) 里
12    ll m = (ll) sqrt((db) l);
13    while (m * m < l) ++m;
14    for(ll y = 0, w = 1 % p; y < std::min(m, l); ++y) { // y < m 的小解
15        if(w == b) return y;
16        w = (i128) w * a % p;
17    }
18    // 大解写成 y = i * m + j (i >= 1, 0 <= j < m): a^j = b * a^(-i * m), 查表中即得
19    unordered_map<ll, ll> pos;
20    pos.reserve(m + 2);
21    for(ll j = 0, w = 1 % p; j < m; ++j) {
22        pos.emplace(w, j);
23        w = (i128) w * a % p;
24    }
25    ll step = ksm(a, m, p), iv = ksm(step, p - 2, p), w = (i128) b * iv % p, ans =
-1;
26    for(ll i = 1; i <= m; ++i, w = (i128) w * iv % p)
27        if(auto it = pos.find(w); it != pos.end()) {
28            ll y = (ll) i * m + it->second;
29            if(ksm(a, y, p) == b && (ans < 0 || y < ans)) ans = y; // 显式验证, 稳一手
30        }
31    return ans < 0 ? -1 : ans;
32 }

```

5.4 中国剩余定理

```

1 ll CRT(ll a1, ll p1, ll a2, ll p2) {
2     ll x, y, g = exgcd(p1, p2, x, y);
3     if((a2 - a1) % g) return -1;
4     ll m1 = p1 / g, m2 = p2 / g;
5     // p1 * t ≡ a2 - a1 (mod p2) 约去 g 后: p1 / g * t ≡ (a2 - a1) / g (mod m2)
6     ll k = (a2 - a1) / g % m2 * (x % m2 + m2) % m2;

```

```

7     ll ans = ((i128) k * p1 + a1) % (m1 * p2); // mod lcm 得到 [0, lcm) 内的解
8     return (ans + m1 * p2) % (m1 * p2);
9 }
10 // 多方程: ans = 0, mod = 1;
11 // For(i, 1, n) {
12 //     ans = CRT(ans, mod, a[i], p[i]);
13 //     if(ans < 0) 无解;
14 //     mod = mod / gcd(mod, p[i]) * p[i];
15 // }

```

5.5 Miller-Rabin 素性检验

```

1 namespace MR {
2     u64 mul(u64 a, u64 b, u64 p) { return (unsigned __int128) a * b % p; }
3     u64 pw(u64 a, u64 x, u64 p, u64 r = 1) {
4         for(; x >= 1, a = mul(a, a, p))
5             if(x & 1) r = mul(r, a, p);
6         return r;
7     }
8 }
9 // 底数取前 12 个素数: 对 u64 范围是确定性判定, 没有错误率。
10 bool chkp(u64 p) {
11     if(p < 2) return 0;
12     if(p < 4) return 1;
13     for(u64 d : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37})
14         if(p % d == 0) return p == d;
15     u64 s = __builtin_ctzll(p - 1), d = p - 1 >> s;
16     for(u64 a : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37}) {
17         u64 x = MR::pw(a, d, p);
18         if(x == 1 || x == p - 1) continue;
19         bool ok = 0;
20         For(i, 1, s - 1) {
21             x = MR::mul(x, x, p);
22             if(x == p - 1) { ok = 1; break; }
23         }
24         if(!ok) return 0;
25     }
26     return 1;
27 }

```

5.6 Pollard-Rho 分解质因数

```

1 u64 isqrt(u64 x) {
2     if(x < 2) return x;
3     u64 r = sqrtl((long double) x);
4     for(; (unsigned __int128) r * r > x; --r);
5     for(; (unsigned __int128) (r + 1) * (r + 1) <= x; ++r);
6     return r;
7 }
8 // 返回 x 的一个非平凡因子; 依赖 chkp (即「Miller-Rabin」模板)。
9 // 先摘掉小因子与完全平方数: 这两种情形会让 rho 只给出平凡因子、空转。
10 u64 rho(u64 x) {

```

```

11     if(x % 2 == 0) return 2;
12     if(x % 3 == 0) return 3;
13     For(d, 5, 100) if(x % d == 0) return d;
14     if(u64 s = isqrt(x); s * s == x) return s;
15     for(; !chkp(x);) {
16         u64 c = 1 + rand() % x, t = 0, r = 0, p = 1;
17         auto f = [&](u64 v) { return ((unsigned __int128) v * v + c) % x; };
18         for(; p == 1; p = gcd(t > r ? t - r : r - t, x)) t = f(t), r = f(f(r));
19         if(p != x) return p; // 平凡因子:换个 c 重来
20     }
21     return x;
22 }
23 // 把 x 的质因子按从小到大存入 res(带重数)。期望复杂度  $O(x^{1/4})$ 。
24 void fact(u64 x, vect<u64> &res) {
25     if(x == 1) return;
26     if(chkp(x)) return res.push_back(x), void();
27     u64 d = rho(x);
28     fact(d, res), fact(x / d, res);
29 }

```

5.7 扩展欧几里得

```

1 ll exgcd(ll a, ll b, ll &x, ll &y) {
2     if(!b) return x = 1, y = 0, a;
3     ll g = exgcd(b, a % b, y, x);
4     return y -= a / b * x, g;
5 }
6 // 求 a 在模 p 下的逆元(要求 gcd(a, p) = 1):
7 // ll x, y; exgcd(a, p, x, y); x = (x % p + p) % p;

```

5.8 最小模线性值

```

1 int minmod(int n, int m, int a, ll b) {
2     int ans = m;
3     for(; n; swap(a, m)) {
4         a %= m, b %= m;
5         if(b < 0) b += m;
6         cmin(ans, (int) b);
7         n = ((ll) (n - 1) * a + b) / m;
8         b -= (ll) m * n;
9     }
10    return ans;
11 }

```

5.9 二次剩余

```

1 bool quadres(u64 a, u64 p) {
2     return !a || ksm(a, p - 1 >> 1, p) == 1;
3 }
4 // sqrt(x, p): 求 x 在模奇素数 p 下的平方根(要求  $0 \leq x < p$ )。
5 // p 为模 4 余 3 的素数时直接算, 否则用 Cipolla; x 非二次剩余时返回 -1。
6 int sqrt(u64 x, ll p) {

```

```

7     if(x <= 1) return x;
8     if(ksm(x, p - 1 >> 1, p) != 1) return -1;
9     if(p % 4 == 3) return ksm(x, p + 1 >> 2, p);
10    static mt19937_64 gen;
11    ll w, a;
12    do a = gen() % p; while (!a || ksm(w = (a * a - x % p + p) % p, p - 1 >> 1, p) != p - 1);
13    auto prod = [&](pll u, pll v) {
14        return make_pair((u.first * v.first + u.second * v.second % p * w) % p,
15            (u.first * v.second + u.second * v.first) % p);
16    };
17    pll r{1, 0}, b{a, 1};
18    for(ll k = p + 1 >> 1; k; k >>= 1, b = prod(b, b))
19        if(k & 1) r = prod(r, b);
20    return r.first;
21 }

```

5.10 原根与阶

```

1 int root(int p) {
2     if(p == 2) return 1;
3     static vect<int> fac;
4     fac = {}, fac.reserve(20);
5     int x = p - 1;
6     for(int i = 2; i * i <= x; ++i)
7         if(x % i == 0) {
8             fac.push_back(i);
9             while (x % i == 0) x /= i;
10        }
11    if(x > 1) fac.push_back(x);
12    For(g, 2, p - 1) {
13        bool ok = 1;
14        for(auto q : fac) ok &= ksm(g, (p - 1) / q, p) != 1;
15        if(ok) return g;
16    }
17    return -1;
18 }
19 // ord(a, p): a 模素数 p 的阶, 即最小的  $d > 0$  使  $a^d \equiv 1 \pmod{p}$ 。要求  $\gcd(a, p) = 1$ 。
20 // 从  $p - 1$  出发, 对  $p - 1$  的每个素因子 q 反复试除(能除动说明阶里不含 q),
21 // 除到除不动为止, 剩下的就是阶。
22 int ord(int a, int p) {
23     if(p == 2) return 1;
24     int x = p - 1;
25     static vect<int> fac;
26     int y = x;
27     fac = {}, fac.reserve(20);
28     for(int i = 2; i * i <= y; ++i)
29         if(y % i == 0) {
30             fac.push_back(i);
31             while (y % i == 0) y /= i;
32         }
33     if(y > 1) fac.push_back(y);

```

```

34     for(auto q : fac)
35         while (x % q == 0 && ksm(a, x / q, p) == 1) x /= q;
36     return x;
37 }

```

5.11 杜教筛

```

1 unordered_map<ll, ll> smu_big, sphi_big;
2 ll DJS_mu(ll n) { // f = mu, 取 g = 1, 则 H(n) = 1
3     if(n <= N) return smu[n];
4     if(smu_big.count(n)) return smu_big[n];
5     ll ans = 1;
6     for(ll l = 2, r; l <= n; l = r + 1)
7         r = n / (n / l), ans -= (r - l + 1) * DJS_mu(n / l);
8     return smu_big[n] = ans;
9 }
10 ll DJS_phi(ll n) { // f = phi, 取 g = 1, 则 H(n) = n * (n + 1) / 2
11     if(n <= N) return sphi[n];
12     if(sphi_big.count(n)) return sphi_big[n];
13     ll ans = (ll) n * (n + 1) / 2; // 先把 n 转成 ll, 否则 n 到 1e6 就溢出
14     for(ll l = 2, r; l <= n; l = r + 1)
15         r = n / (n / l), ans -= (r - l + 1) * DJS_phi(n / l);
16     return sphi_big[n] = ans;
17 }
18 // 通用形式: 只需实现 f 与 g, 其余照抄上面两个函数的结构
19 // ll S(ll n) {
20 //     if(n <= N) return pre[n];
21 //     if(mp.count(n)) return mp[n];
22 //     ll ans = H(n); // f * g 的前缀和
23 //     for(ll l = 2, r; l <= n; l = r + 1)
24 //         r = n / (n / l), ans -= (g_pre(r) - g_pre(l - 1)) * S(n / l);
25 //     return mp[n] = ans;
26 // }

```

5.12 线性筛

```

1 struct sieve_t {
2     int p[N / 10 + 5], tot, mn[N + 1], phi[N + 1], mu[N + 1], d[N + 1], cnt[N + 1],
3     lpf[N + 1];
4     ll s[N + 1];
5     void work() {
6         tot = 0, phi[1] = mu[1] = d[1] = s[1] = 1;
7         For(i, 2, N) {
8             if(!mn[i])
9                 p[++tot] = i, mn[i] = i, phi[i] = i - 1, mu[i] = -1,
10                 d[i] = 2, cnt[i] = 1, s[i] = i + 1, lpf[i] = i + 1;
11             for(int j = 1, x; j <= tot && (x = i * p[j]) <= N; ++j) {
12                 mn[x] = p[j], cnt[x] = 1, lpf[x] = p[j] + 1;
13                 if(i % p[j] == 0) {
14                     phi[x] = phi[i] * p[j], mu[x] = 0;
15                     cnt[x] = cnt[i] + 1, lpf[x] = lpf[i] * p[j] + 1;
16                     d[x] = d[i] / (cnt[i] + 1) * (cnt[x] + 1);

```

```

16         s[x] = s[i] / lpf[i] * lpf[x];
17         break;
18     }
19     phi[x] = phi[i] * (p[j] - 1), mu[x] = -mu[i];
20     d[x] = d[i] * 2, s[x] = s[i] * (p[j] + 1);
21 }
22 }
23 }
24 bool isp(int x) const { return x >= 2 && mn[x] == x; }
25 } S;
26 // 莫比乌斯/欧拉函数的前缀和(杜教筛要用): 筛完后
27 // For(i, 1, N) smu[i] = smu[i - 1] + S.mu[i], sphi[i] = sphi[i - 1] + S.phi[i];

```

5.13 下取整和

求 $\sum_{i=0}^{n-1} \lfloor (a \cdot i + b) / m \rfloor$, 复杂度 $O(\log m)$ 。参数全是无符号, a, b 不必小于 m 。

用法

- 直接调用: `floor_sum(n, m, a, b)`, 返回上面那个和; $n = 0$ 时返回 0。
- 求的是从 $i = 0$ 起的 n 项。若题目是 $\sum_{i=1}^n$, 写成 `floor_sum(n + 1, m, a, b)` 或先扣掉 $i = 0$ 的 b / m 。
- 从 l 到 r 的区间和: `floor_sum(r + 1, m, a, b) - floor_sum(l, m, a, b)`。
- 需要 $\sum_{i=0}^{n-1} (a \cdot i + b)$ 这类“上取整/整除”时, 先化成下取整再套本模板。

契约与坑

- 答案本身不能超过 $2^{64} - 1$ (累加器是 `u64`, 只有中间量 y 用了 `i128`): 大致 $n^2 \cdot a / m$ 量级, $n \leq 1e9$ 、 $a < m \leq 1e9$ 时安全; 更大的话把返回类型与 `ans` 一并换成 `unsigned __int128`。
- 参数是无符号的, 别传负数(负数下取整得自己先按 m 归正)。
- 它不等于数学章那个「类欧」: 类欧.cpp 是求 $\lfloor (ai+b)/c \rfloor$ 的 $f/g/h$ 三个量的递归模板, 本模板只求那一项下取整和。

```

1 u64 floor_sum(u64 n, u64 m, u64 a, u64 b) {
2     u64 ans = 0;
3     for(;;) {
4         if(a >= m) ans += n * (n - 1) / 2 * (a / m), a %= m;
5         if(b >= m) ans += n * (b / m), b %= m;
6         u64 y = (i128) a * n + b;
7         if(y < m) break;
8         n = y / m, b = y % m, swap(m, a);
9     }
10    return ans;
11 }

```

6 多项式

6.1 任意模数 NTT

```
1 namespace Mul {
2 using poly = vector<int>;
3 template <int M, int G> static void NTT(poly &a, const int k) {
4     auto ksm = [&](ll a, int b) {ll
5         ans=1;for(;b>=1;a=a%M)if(b&1)ans=ans*a%M;return ans; };
6     static poly r, w;
7     int n = a.size();
8     if(n != r.size()) {
9         int l = __lg(n);
10        r.resize(n);
11        For(i, 0, n - 1) r[i] = (r[i >> 1] >> 1) | ((i & 1) << (l - 1));
12    }
13    For(i, 0, n - 1) if(i < r[i]) swap(a[r[i]], a[i]);
14    ll wi = ksm(G, (M - 1 >> 1) / n * k + M - 1);
15    w.resize(n), w[0] = 1;
16    For(i, 1, n - 1) w[i] = w[i - 1] * wi % M;
17    for(int m = 1, l = __lg(n); m < n; m <= 1, --l)
18        for(int i = 0; i < n; i += m + m)
19            for(j, 0, m - 1) {
20                ll x = a[i + j], y = 1ll * w[j << l] * a[i + j + m] % M;
21                a[i + j] = x + y >= M ? x + y - M : x + y;
22                a[i + j + m] = x - y < 0 ? x - y + M : x - y;
23            }
24    if(k == -1) {
25        ll inv = ksm(n, M - 2);
26        For(i, 0, n - 1) a[i] = a[i] * inv % M;
27    }
28 poly mul(poly a, poly b, int P) {
29     int n = a.size(), m = b.size();
30     if(!n || !m) return poly();
31     int l = 1 << __lg(n + m - 2) + 1;
32     a.resize(l), b.resize(l);
33     static constexpr int M[3]{998244353, 1004535809, 469762049}, G = 3;
34 #define g(p) \
35     [&](poly a, poly b) { \
36         NTT<M[p], G>(a, 1), NTT<M[p], G>(b, 1); \
37         For(i, 0, l - 1) a[i] = 1ll * a[i] * b[i] % M[p]; \
38         NTT<M[p], G>(a, -1); \
39         return a; \
40     }(a, b)
41     poly c[3]{g(0), g(1), g(2)}, d;
42     d.reserve(n + m - 1);
43 #undef g
44     static constexpr ll m01 = 1ll * M[0] * M[1];
45     static constexpr int i0 = 669690699, i1 = 354521948;
46     For(id, 0, n + m - 2) {
47         auto A = c[0][id], B = c[1][id], C = c[2][id];
48         ll x = 1ll * (B - A + M[1]) % M[1] * i0 % M[1] * M[0] + A;
```

```
49         d.push_back((1ll * (C - x % M[2] + M[2]) % M[2] * i1 % M[2] * (m01 % P) % P
50         + x) % P);
51     }
52     return d;
53 } // namespace Mul
54 using poly = vector<mint>;
55 int M;
56 poly mul(poly a, const poly &b) {
57     int n = a.size(), m = b.size();
58     if(!n || !m) return poly();
59     vector<int> c(n), d(m);
60     For(i, 0, n - 1) c[i] = a[i].x;
61     For(i, 0, m - 1) d[i] = b[i].x;
62     c = Mul::mul(c, d, M);
63     a.resize(c.size());
64     For(i, 0, a.size() - 1) a[i] = c[i];
65     return a;
66 }
```

6.2 多项式复合

模 998244353, $F(G(x)) \bmod x^n$, 一次调用全部解出。

- 完整程序(自带 main): 第一行 n , 随后两行系数; 输出复合结果 n 个系数
- $\text{comp}(F, G, n)$: 接口, 要求 $[x^0]G = 0$; $\text{prep}(22)$ 一次性预处理 NTT 单位根(支持长度 $\leq 2^{22}$)
- 内部 $\text{solve}(F, G, n, m)$: 分治卷积, 先乘 $A \cdot B$ 再做奇偶抽取
- 复杂度: 多项式级(小常数大), 适合 $n \leq 10^5$ 量级

坑: 数组全局且巨大($4.4e6 * \text{十几个}$), 内存约 300+ MB, 跑题前确认空间; $\text{scanf}/\text{printf}$ 输入输出; F 与 G 都从常数项开始读入。

```
1 #include <bits/stdc++.h>
2
3 using namespace std;
4 typedef vector<int> vi;
5 typedef vector<vi> vv;
6 typedef long long ll;
7 typedef unsigned long long ull;
8 const int N = 4.4e6 + 15, mod = 998244353;
9 int F[N], X[N], Y[N], W[N];
10 ull Z[N];
11 int rev[N * 2], omg[N * 2];
12 int qpow(int x, int y)
13 {
14     int ans = 1;
15     while(y)
16     {
17         if(y & 1)
18             ans = (ll)ans * x % mod;
19         x = (ll)x * x % mod;
20         y >>= 1;
21     }
```

```

22     return ans;
23 }
24 void prep(int n)
25 {
26     int i, j, w;
27     for(i = 0; i <= n; i++)
28     {
29         for(j = 1; j < (1 << i); j++)
30             rev[(1 << i) + j] = rev[(1 << (i - 1)) + (j >> 1)] | ((j & 1) << (i - 1));
31         w = qpow(3, (mod - 1) >> i);
32         omg[1 << i] = 1;
33         for(j = 1; j < (1 << i); j++)
34             omg[(1 << i) + j] = (ll)omg[(1 << i) + j - 1] * w % mod;
35     }
36 }
37 ull tmp[N * 2];
38 inline void ntt(int *A, int n, bool t) // 0 DFT 1 IDFT
39 {
40     int i, j, k, q, x;
41     for(i = 0; i < (1 << n); i++)
42         tmp[i] = A[i];
43     for(i = 0; i < (1 << n); i++)
44         if(rev[(1 << n) + i] > i)
45             swap(tmp[rev[(1 << n) + i]], tmp[i]);
46     for(i = 0; i < n; i++)
47     {
48         for(j = 0; j < (1 << n); j += (2 << i))
49             for(k = 0; k < (1 << i); k++)
50             {
51                 x = tmp[j + k + (1 << i)] * omg[(2 << i) + k] % mod;
52                 tmp[j + k + (1 << i)] = tmp[j + k] + mod - x;
53                 tmp[j + k] = tmp[j + k] + x;
54             }
55         if(i == 17 || i == n - 1)
56             for(j = 0; j < (1 << n); j++)
57                 tmp[j] %= mod;
58     }
59     for(i = 0; i < (1 << n); i++)
60         A[i] = tmp[i];
61     if(t)
62     {
63         for(i = 1; i < (1 << n); i++)
64             if(i < (1 << n) - i)
65                 swap(A[i], A[(1 << n) - i]);
66         q = qpow(1 << n, mod - 2);
67         for(i = 0; i < (1 << n); i++)
68             A[i] = (ll)A[i] * q % mod;
69     }
70 }
71 struct poly
72 {
73     vector<int> c;
74     poly()

```

```

75     {
76         c.clear();
77     }
78     poly(vector<int> d)
79     {
80         c = d;
81     }
82     friend poly operator * (poly f, poly g)
83     {
84         if(f.c.empty() || g.c.empty())
85             return poly();
86         poly h;
87         int i, l = 0;
88         while((f.c.size() + g.c.size()) >> l)
89             l++;
90         memset(X, 0, 4 << l);
91         memset(Y, 0, 4 << l);
92         for(i = 0; i < f.c.size(); i++)
93             X[i] = f.c[i];
94         for(i = 0; i < g.c.size(); i++)
95             Y[i] = g.c[i];
96         ntt(X, l, 0);
97         ntt(Y, l, 0);
98         for(i = 0; i < (1 << l); i++)
99             X[i] = (ll)X[i] * Y[i] % mod;
100         ntt(X, l, 1);
101         h.c.resize(f.c.size() + g.c.size() - 1);
102         for(i = 0; i < h.c.size(); i++)
103             h.c[i] = X[i];
104         return h;
105     }
106 };
107 vv solve(vi F, vv G, int n, int m) // [x^[0, n)y^(-m, 0]], [x^0]G = 1
108 {
109     int i, j, S, T, a, b, d = 2 * n - 1;
110     vv ans;
111     if(n == 1)
112     {
113         F.resize(m);
114         ans.resize(m);
115         for(i = 0; i < m; i++)
116             ans[i] = {F[i]};
117         return ans;
118     }
119     poly A, B, C;
120     vv H;
121     S = m * d + n;
122     A.c.resize(S);
123     B.c.resize(S);
124     for(i = 0; i <= m; i++)
125         for(j = 0; j < n; j++)
126         {
127             A.c[i * d + j] = G[i][j];

```

```

128     B.c[i * d + j] = (j & 1 ? mod - G[i][j] : G[i][j]);
129 }
130 A = A * B;
131 a = (n + 1) / 2;
132 b = 2 * m;
133 H.resize(b + 1);
134 for(i = 0; i <= b; i++)
135 {
136     H[i].resize(a);
137     for(j = 0; j < a; j++)
138         H[i][j] = A.c[i * d + 2 * j];
139 }
140 auto P = solve(F, H, a, b);
141 T = (b - 1) * d + n;
142 C.c.resize(T);
143 for(i = 0; i < b; i++)
144     for(j = 0; j < a; j++)
145         C.c[i * d + 2 * j] = P[b - 1 - i][j];
146 C = C * B;
147 ans.resize(m);
148 for(i = 0; i < m; i++)
149 {
150     ans[i].resize(n);
151     for(j = 0; j < n; j++)
152         ans[i][j] = C.c[(b - 1 - i) * d + j];
153 }
154 return ans;
155 }
156 poly comp(poly F, poly G, int n) // [x^0]G = 0
157 {
158     int i;
159     vv Gp;
160     Gp.resize(2);
161     Gp[0].resize(n);
162     Gp[1].resize(n);
163     Gp[0][0] = 1;
164     for(i = 1; i < n; i++)
165         Gp[1][i] = mod - G.c[i];
166     auto P = solve(F.c, Gp, n, 1);
167     return poly(P[0]);
168 }
169 int main()
170 {
171     prep(22);
172     int n, i;
173     poly F, G;
174     scanf("%d", &n);
175     F.c.resize(n);
176     G.c.resize(n);
177     for(i = 0; i < n; i++)
178         scanf("%d", &F.c[i]);
179     for(i = 0; i < n; i++)
180         scanf("%d", &G.c[i]);

```

```

181     auto H = comp(F, G, n);
182     for(i = 0; i < n; i++)
183         printf("%d ", H.c[i]);
184     printf("\n");
185     return 0;
186 }

```

6.3 多项式复合逆

模 998244353, 给定 F , 求逆 G 使 $F(G(x)) = x \pmod{x^n}$ 。

- 完整程序(自带 main): 输入 n 与 F 的 n 个系数, 输出 G 的 n 个系数
- comp_inv(F, n): 算法基于 Lagrange 反演 $[x^n] F^{-k}$, 内部 pw_pj 做形式幂级数逆 + 幂迭代
- 要求: $[x^0] F = 0$ 且 $[x^1] F \neq 0$ (首项必须可逆, 代码里直接求 qpow(F[1], mod-2))

坑: 与「多项式复合」同一套全局数组体系(数组大小约 $4.4e6+10$), 内存大, 搬用注意符号冲突; prep(22) 除单位根外还预处理好阶乘/逆元表(fac/inv/Inv), 整个程序只调用一次。

```

1 #include <bits/stdc++.h>
2
3 using namespace std;
4 typedef vector<int> vi;
5 typedef vector<vi> vv;
6 typedef long long ll;
7 typedef unsigned long long ull;
8 const int N = 4.4e6 + 15, mod = 998244353;
9 int F[N], X[N], fac[N], inv[N], Inv[N];
10 int rev[N * 2], omg[N * 2];
11 int qpow(int x, int y)
12 {
13     int ans = 1;
14     while(y)
15     {
16         if(y & 1)
17             ans = (ll)ans * x % mod;
18         x = (ll)x * x % mod;
19         y >>= 1;
20     }
21     return ans;
22 }
23 void prep(int n)
24 {
25     int i, j, w, m = N - 5;
26     for(i = 0; i <= n; i++)
27     {
28         for(j = 1; j < (1 << i); j++)
29             rev[(1 << i) + j] = rev[(1 << (i - 1)) + (j >> 1)] | ((j & 1) << (i - 1));
30         w = qpow(3, (mod - 1) >> i);
31         omg[1 << i] = 1;
32         for(j = 1; j < (1 << i); j++)
33             omg[(1 << i) + j] = (ll)omg[(1 << i) + j - 1] * w % mod;
34     }
35     fac[0] = 1;

```

```

36 for(i = 1; i <= m; i++)
37     fac[i] = (ll)fac[i - 1] * i % mod;
38 inv[m] = qpow(fac[m], mod - 2);
39 for(i = m; i >= 1; i--)
40     inv[i - 1] = (ll)inv[i] * i % mod;
41 for(i = 1; i <= m; i++)
42     Inv[i] = (ll)inv[i] * fac[i - 1] % mod;
43 }
44 ull tmp[N * 2];
45 inline void ntt(int *A, int n, bool t) // 0 DFT 1 IDFT
46 {
47     int i, j, k, q, x;
48     for(i = 0; i < (1 << n); i++)
49         tmp[i] = A[i];
50     for(i = 0; i < (1 << n); i++)
51         if(rev[(1 << n) + i] > i)
52             swap(tmp[rev[(1 << n) + i]], tmp[i]);
53     for(i = 0; i < n; i++)
54     {
55         for(j = 0; j < (1 << n); j += (2 << i))
56             for(k = 0; k < (1 << i); k++)
57             {
58                 x = tmp[j + k + (1 << i)] * omg[(2 << i) + k] % mod;
59                 tmp[j + k + (1 << i)] = tmp[j + k] + mod - x;
60                 tmp[j + k] = tmp[j + k] + x;
61             }
62         if(i == 17 || i == n - 1)
63             for(j = 0; j < (1 << n); j++)
64                 tmp[j] %= mod;
65     }
66     for(i = 0; i < (1 << n); i++)
67         A[i] = tmp[i];
68     if(t)
69     {
70         for(i = 1; i < (1 << n); i++)
71             if(i < (1 << n) - i)
72                 swap(A[i], A[(1 << n) - i]);
73         q = qpow(1 << n, mod - 2);
74         for(i = 0; i < (1 << n); i++)
75             A[i] = (ll)A[i] * q % mod;
76     }
77 }
78 vi pv(vi F, int n, bool t)
79 {
80     int i;
81     F.resize(1 << n);
82     for(i = 0; i < (1 << n); i++)
83         X[i] = F[i];
84     ntt(X, n, t);
85     for(i = 0; i < (1 << n); i++)
86         F[i] = X[i];
87     return F;
88 }

```

```

89 struct poly
90 {
91     vector<int> c;
92     poly()
93     {
94         c.clear();
95     }
96     poly(vector<int> d)
97     {
98         c = d;
99     }
100 friend poly operator * (poly f, poly g)
101 {
102     if(f.c.empty() || g.c.empty())
103         return poly();
104     poly h;
105     int i, l = 0;
106     while((f.c.size() + g.c.size()) >> l)
107         l++;
108     f.c = pv(f.c, l, 0);
109     g.c = pv(g.c, l, 0);
110     for(i = 0; i < (1 << l); i++)
111         X[i] = (ll)f.c[i] * g.c[i] % mod;
112     ntt(X, l, 1);
113     h.c.resize(f.c.size() + g.c.size() - 1);
114     for(i = 0; i < h.c.size(); i++)
115         h.c[i] = X[i];
116     return h;
117 }
118 };
119 poly pw_pj(poly F, int n) // [x^{n-1}]1/(1-yF)
120 {
121     int i, j, S, R, m = 1, d = n - 1, l, r;
122     vv A, B;
123     vi pA, pB, pC;
124     A.resize(2);
125     A[0].resize(n);
126     A[0][0] = 1;
127     A[1].resize(n);
128     B = A;
129     for(i = 0; i < n; i++)
130         B[1][i] = mod - F.c[i];
131     while(d)
132     {
133         R = 2 * d + 1;
134         S = m * R + d + 1;
135         l = 0;
136         while((S - 1) >> l)
137             l++;
138         l++;
139         pA = pB = pC = {};
140         pA.resize(S);
141         pB.resize(S);

```

```

142 pC.resize(S);
143 for(i = 0; i <= m; i++)
144     for(j = 0; j <= d; j++)
145     {
146         pA[i * R + j] = A[i][j];
147         pB[i * R + j] = B[i][j];
148         pC[i * R + j] = (j & 1 ? mod - B[i][j] : B[i][j]);
149     }
150 pA = pv(pA, l, 0);
151 pB = pv(pB, l, 0);
152 pC = pv(pC, l, 0);
153 for(i = 0; i < (1 << l); i++)
154 {
155     pA[i] = (ll)pA[i] * pC[i] % mod;
156     pB[i] = (ll)pB[i] * pC[i] % mod;
157 }
158 pA = pv(pA, l, 1);
159 pB = pv(pB, l, 1);
160 r = d & 1;
161 d /= 2;
162 m *= 2;
163 A.resize(m + 1);
164 B.resize(m + 1);
165 for(i = 0; i <= m; i++)
166 {
167     A[i].resize(d + 1);
168     B[i].resize(d + 1);
169     for(j = 0; j <= d; j++)
170     {
171         A[i][j] = pA[i * R + j * 2 + r];
172         B[i][j] = pB[i * R + j * 2];
173     }
174 }
175 }
176 for(i = 0; i < n; i++)
177     F.c[i] = A[i][0];
178 return F;
179 }
180 poly comp_inv(poly F, int n) // [x^n]F^k=k/n[x^{n-k}](G/x)^{-n}
181 {
182     int i, r, p = qpow(F.c[1], mod - 2), c;
183     for(i = 0, c = 1; i < n; i++, c = (ll)c * p % mod)
184         F.c[i] = (ll)F.c[i] * c % mod;
185     poly G;
186     F = pw_pj(F, n);
187     reverse(F.c.begin(), F.c.end());
188     for(i = 0; i < n; i++)
189         F.c[i] = (ll)F.c[i] * Inv[n - 1 - i] % mod * (n - 1) % mod;
190     r = mod - Inv[n - 1];
191     G.c = {1};
192     while(r)
193     {
194         if(r & 1)

```

```

195     {
196         G = G * F;
197         G.c.resize(n);
198     }
199     F = F * F;
200     F.c.resize(n);
201     r >>= 1;
202 }
203 G.c.resize(n);
204 for(i = n - 1; i > 0; i--)
205     G.c[i] = (ll)G.c[i - 1] * p % mod;
206 G.c[0] = 0;
207 return G;
208 }
209 int main()
210 {
211     prep(22);
212     int n, i;
213     scanf("%d", &n);
214     poly F;
215     F.c.resize(n);
216     for(i = 0; i < n; i++)
217         scanf("%d", &F.c[i]);
218     F = comp_inv(F, n);
219     for(i = 0; i < n; i++)
220         printf("%d ", F.c[i]);
221     printf("\n");
222     return 0;
223 }

```

7 计算几何

namespace Geo,eps=1e-10,双精度;p2 为点,seg 为线段。

- 基本: sign/cmp,det(叉积)、点乘 *、r90/rot/unit、dis/dis2
- 线段: chkss(相交含端点)/chkss_s(严格)、isll(直线交)、ons/ons_s(点在线上)、proj/reflect/nearest/diss(投影/翻转/点线距/线线距)
- 多边形: area(面积,有符号)、contain(点形关系: 0 外/1 线上/2 内)、convex_hull(凸包,默认去共线点,nos=1 保留)、convex_diameter(凸包直径)、convex_cut(半平面裁剪)
- 圆: circumcenter(外心)、circumcircle_diameter(外接圆直径,三点共线返 -1)

坑: operator-= 和 operator/= 实现成了 $x = x + y$ (笔误!)——需要减/除时手写;area 输出带符号(按顶点序),要面积就取 abs;convex_hull 会改写输入数组,保序请另存;eps=1e-10 对边界情况(多点共线等)需要留心。

```

1 namespace Geo {
2
3 const db eps = 1e-10;
4
5 struct p2;
6 struct seg;
7 int sign(db x) { return x < -eps ? -1 : x > eps; }

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8 int cmp(db x, db y) { return sign(x - y); }
9 const db pi = acos(db(-1));
10 struct p2 {
11     db x, y;
12     db det(const p2 &b) const { return x * b.y - y * b.x; }
13     db alpha() const { return atan2(y, x); }
14     bool operator==(const p2 &b) const {
15         return !cmp(x, b.x) && !cmp(y, b.y);
16     }
17     bool operator<(const p2 &b) const {
18         int c = cmp(x, b.x);
19         if(c) return !~c;
20         return !~cmp(y, b.y);
21     }
22 };
23 p2 r90(p2 x) { return {-x.y, x.x}; }
24 p2 rot(p2 x, db a) { return {x.x * cos(a) - x.y * sin(a), x.x * sin(a) + x.y *
    cos(a)}; }
25 p2 operator+(p2 x, p2 y) { return {x.x + y.x, x.y + y.y}; }
26 p2 operator-(p2 x, p2 y) { return {x.x - y.x, x.y - y.y}; }
27 p2 operator/(p2 x, db y) { return {x.x / y, x.y / y}; }
28 p2 operator*(p2 x, db y) { return {x.x * y, x.y * y}; }
29 p2 operator*(db y, p2 x) { return {x.x * y, x.y * y}; }
30 db operator*(p2 x, p2 y) { return x.x * y.x + x.y * y.y; }
31 p2 operator+=(p2 &x, p2 y) { return x = x + y; }
32 p2 operator-=(p2 &x, p2 y) { return x = x - y; }
33 p2 operator/=(p2 &x, p2 y) { return x = x / y; }
34 db dis2(p2 x) { return x.x * x.x + x.y * x.y; }
35 db dis(p2 x) { return hypot(x.x, x.y); }
36 p2 unit(p2 x) { return x / dis(x); }
37 struct seg {
38     p2 x, y;
39     p2 dir() const { return y - x; }
40 };
41 db cross(const seg &p, const p2 &q) { return p.dir().det(q - p.x); }
42 int crossop(const seg &p, const p2 &q) { return sign(cross(p, q)); }
43 bool inc(seg l, const p2 &b) { return !crossop(l, b); }
44 bool eqll(const seg &a, const seg &b) {
45     return inc(a, b.x) && inc(a, b.y);
46 }
47 bool chkll(const seg &a, const seg &b) {
48     db a1 = cross(b, a.x), a2 = -cross(b, a.y);
49     return !!sign(a1 + a2);
50 }
51 p2 isll(const seg &a, const seg &b) {
52     db a1 = cross(b, a.x), a2 = -cross(b, a.y);
53     return (a.x * a2 + a.y * a1) / (a1 + a2);
54 }
55 bool chkss(const seg &p, const seg &q) {
56     auto is = [](db l1, db r1, db l2, db r2) {
57         if(l1 > r1) swap(l1, r1);
58         if(l2 > r2) swap(l2, r2);
59         return !(cmp(r1, l2) == -1 || cmp(r2, l1) == -1);

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60     };
61     p2 p1 = p.x, p2 = p.y, q1 = q.x, q2 = q.y;
62     return is(p1.x, p2.x, q1.x, q2.x) && is(p1.y, p2.y, q1.y, q2.y) &&
63         crossop(p, q1) * crossop(p, q2) <= 0 && crossop(q, p1) * crossop(q, p2)
    <= 0;
64 }
65 bool chkss_s(const seg &p, const seg &q) {
66     p2 p1 = p.x, p2 = p.y, q1 = q.x, q2 = q.y;
67     return crossop(p, q1) * crossop(p, q2) < 0 && crossop(q, p1) * crossop(q, p2) <
    0;
68 }
69 bool isMid(const db &a, const db &m, const db &b) { return !cmp(a, b) || !cmp(b, m)
    || cmp(a, m) != cmp(b, m); }
70 bool isMid(const p2 &a, const p2 &m, const p2 &b) { return isMid(a.x, m.x, b.x) &&
    isMid(a.y, m.y, b.y); }
71 bool ons(const seg &p, const p2 &q) { return !crossop(p, q) && isMid(p.x, q, p.y); }
72 bool ons_s(const seg &p, const p2 &q) {
73     return !crossop(p, q) && sign((q - p.x) * (p.x - p.y)) * sign((q - p.y) * (p.x -
    p.y)) < 0;
74 }
75 p2 proj(const seg &p, const p2 &q) {
76     p2 dir = p.y - p.x;
77     return p.x + dir * ((dir * (q - p.x)) / dis2(dir));
78 }
79 p2 reflect(const seg &p, const p2 &q) {
80     return proj(p, q) * 2 - q;
81 }
82 db nearest(const seg &p, const p2 &q) {
83     p2 h = proj(p, q);
84     if(isMid(p.x, h, p.y)) return dis2(q - h);
85     return min(dis2(p.x - q), dis2(p.y - q));
86 }
87 db disss(const seg &p, const seg &q) {
88     if(chkss(p, q)) return 0;
89     return min({nearest(p, q.x), nearest(p, q.y), nearest(q, p.x), nearest(q,
    p.y)});
90 }
91 db area(int n, p2 *a) {
92     db ret = 0;
93     For(i, 0, n - 1) ret += a[i].det(a[(i + 1) % n]);
94     return ret / 2;
95 }
96 int contain(int n, p2 *a, const p2 &p) { // 0: outside, 1: on seg, 2: inside
97     int ret = 0;
98     For(i, 0, n - 1) {
99         p2 u = a[i], v = a[(i + 1) % n];
100         if(ons({u, v}, p)) return 1;
101         if(cmp(u.y, v.y) <= 0) swap(u, v);
102         if(cmp(p.y, u.y) > 0 || cmp(p.y, v.y) <= 0) continue;
103         ret ^= crossop({p, u}, v) > 0;
104     }
105     return ret * 2;
106 }

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107 int convex_hull(int n, p2 *a, p2 *b, bool nos = 0) {
108     if(n <= 1) return 1;
109     sort(a, a + n);
110     int k = 0;
111     For(i, 0, n - 1) {
112         while(k > 1 && crossop({b[k - 2], b[k - 1]}, a[i]) <= -nos) --k;
113         b[k++] = a[i];
114     }
115     int t = k;
116     rFor(i, n - 2, 0) {
117         while(k > t && crossop({b[k - 2], b[k - 1]}, a[i]) <= -nos) --k;
118         if(i) b[k++] = a[i];
119     }
120     return k;
121 }
122 db convex_diameter(int n, p2 *a) {
123     if(n <= 1) return 0;
124     int is = 0, js = 0;
125     For(k, 1, n - 1) is = a[k] < a[is] ? k : is, js = a[js] < a[k] ? k : js;
126     int i = is, j = js;
127     db ret = dis2(a[i] - a[j]);
128     do {
129         if((a[(i + 1) % n] - a[i]).det(a[(j + 1) % n] - a[j]) >= 0)
130             (++j) %= n;
131         else
132             (++i) %= n;
133         ret = max(ret, dis2(a[i] - a[j]));
134     } while(i != is || j != js);
135     return ret;
136 }
137 vector<p2> convex_cut(int n, p2 *a, const seg &q) {
138     vector<p2> qs;
139     For(i, 0, n - 1) {
140         p2 p1 = a[i], p2 = a[(i + 1) % n];
141         int d1 = crossop(q, p1), d2 = crossop(q, p2);
142         if(d1 >= 0) qs.push_back(p1);
143         if(d1 * d2 < 0) qs.push_back(isll({p1, p2}, q));
144     }
145     return qs;
146 }
147 db circumcircle_diameter(p2 x, p2 y, p2 z) {
148     if(!crossop({x, y}, z)) return -1;
149     return dis(x - y) * dis(y - z) * dis(z - x) / abs((y - x).det(z - x));
150 }
151 array<db, 2> circumcenter(p2 x, p2 y, p2 z) {
152     auto [x1, y1] = x;
153     auto [x2, y2] = y;
154     auto [x3, y3] = z;
155     auto A = dis2(x), B = dis2(y), C = dis2(z);
156     auto u1 = x1 - x2, u2 = x1 - x3, v1 = y1 - y2, v2 = y1 - y3;
157     return {db((C - A) * v1 - (B - A) * v2) / (2 * u1 * v2 - 2 * u2 * v1), db((C -
158     A) * u1 - (B - A) * u2) / (2 * v1 * u2 - 2 * v2 * u1)};
159 }

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159
160 } // namespace Geo
161 using namespace Geo;

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